

Sales, products and customer performance

Business objective. Translate invoice-line records into an interactive view of recorded value, product demand, customer spending and geographic differences, with explicit data-quality limits.

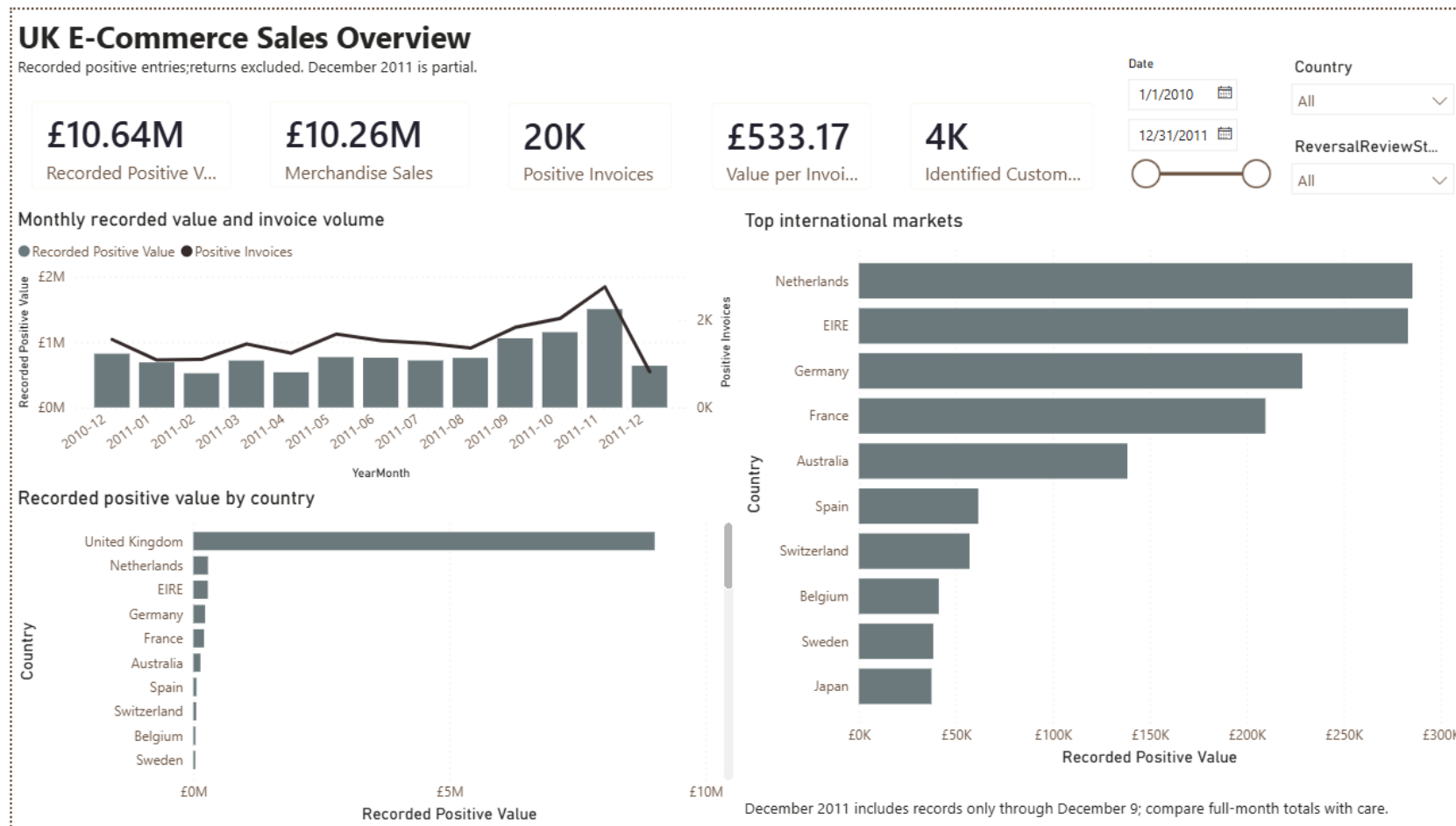
Baseline metric	Result	Definition
Recorded positive value	£10,642,110.80	All retained positive entries; not net revenue
Named-merchandise sales	£10,255,019.18	Classified charges/accounting and MANUAL entries excluded
Merchandise units	5,561,564	Units on named-merchandise positive lines
Positive invoices / invoice lines	19,960 / 524,878	Distinct invoices versus individual records
Identified-customer value	£8,887,208.89	Nonblank CustomerID records
Value per invoice / top-five share	£533.17 / 11.8%	Overall invoice average / identified-customer denominator

Main findings. November 2011 has the highest full-month activity; the UK dominates recorded value; Netherlands leads the displayed qualifying international invoice averages. Two large positive invoices have matching negative entries, materially affecting product and customer rankings.

Scope. Actual coverage: 1 December 2010 to 9 December 2011. Source: UK Online Retail on Kaggle. This is descriptive analysis, with no profit or forecasting claim.

Evidence basis. Dashboard screenshots supplied by the author and reconciled notebook results from this project conversation. The completed PBIX was not inspected. Screenshots are reproduced as supplied; several cards are rounded and labels are clipped. This PDF is a static analytical report, not an interactive export.

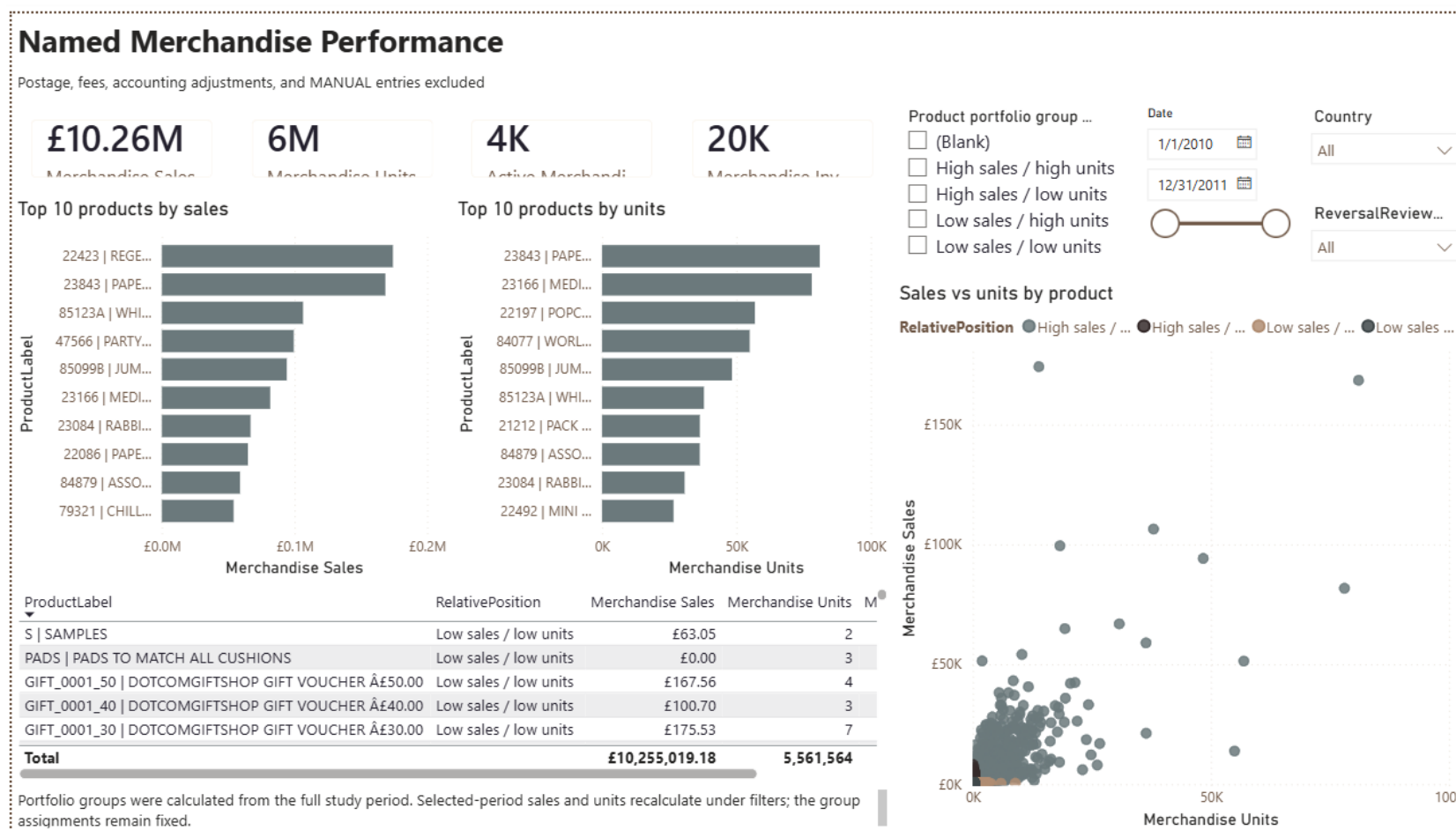
01 | Sales overview



Observed pattern. November is the highest full-month peak. The UK dominates overall value; Netherlands and EIRE lead international recorded value, followed by Germany and France.

Notebook reconciliation. November: £1,503,866.78, up 30.6% from October; invoices rise from 2,040 to 2,769 while value per invoice falls from £564.34 to £543.11. December is partial, so its lower bar is not a like-for-like decline.

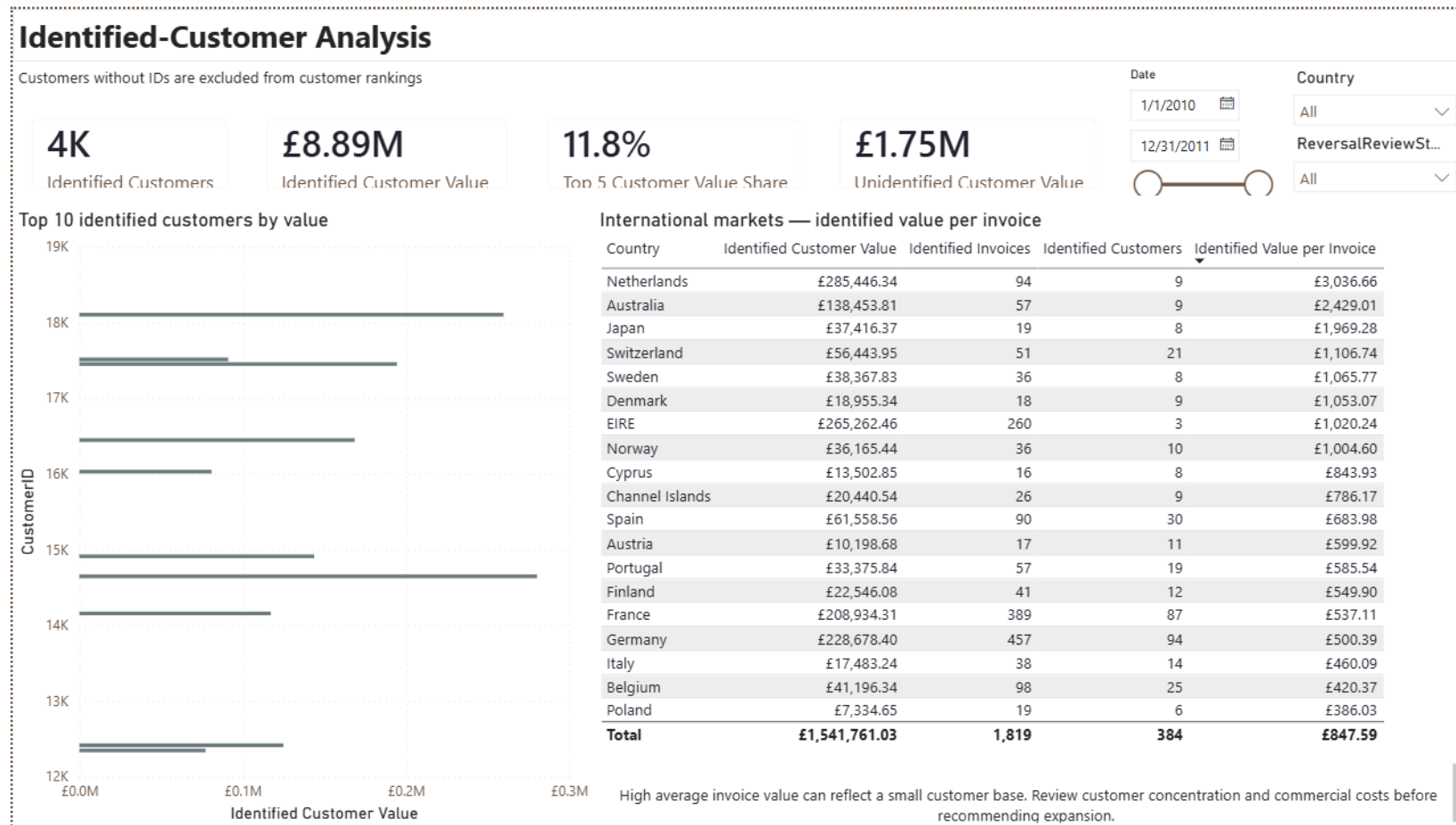
02 | Product performance



Observed pattern. REGENCY CAKESTAND 3 TIER leads sales; PAPER CRAFT, LITTLE BIRDIE leads positive units. The table totals reconcile to £10,255,019.18 and 5,561,564 units.

Interpretation. Little Birdie's large line has a matching negative entry. Its ranking is not evidence of sustained demand. Median portfolio groups are full-period, fixed labels; selected-period sales and units recalculate. Retained vouchers and samples need separate treatment before physical inventory decisions.

03 | Customer and international value



Observed pattern. Identified customers contribute about £8.89m; unassigned value is about £1.75m (16.5% of overall value). Top five share is 11.8% of identified value. Netherlands: £3,036.66 per identified invoice, 94 invoices and nine customers; Australia: £2,429.01; Japan: £1,969.28.

Interpretation. High invoice value can reflect a small customer base. Review margins, repeat ordering and logistics before expansion. The supplied customer chart uses a numeric axis; the text CustomerLabel column documented in the companion file provides a categorical display.

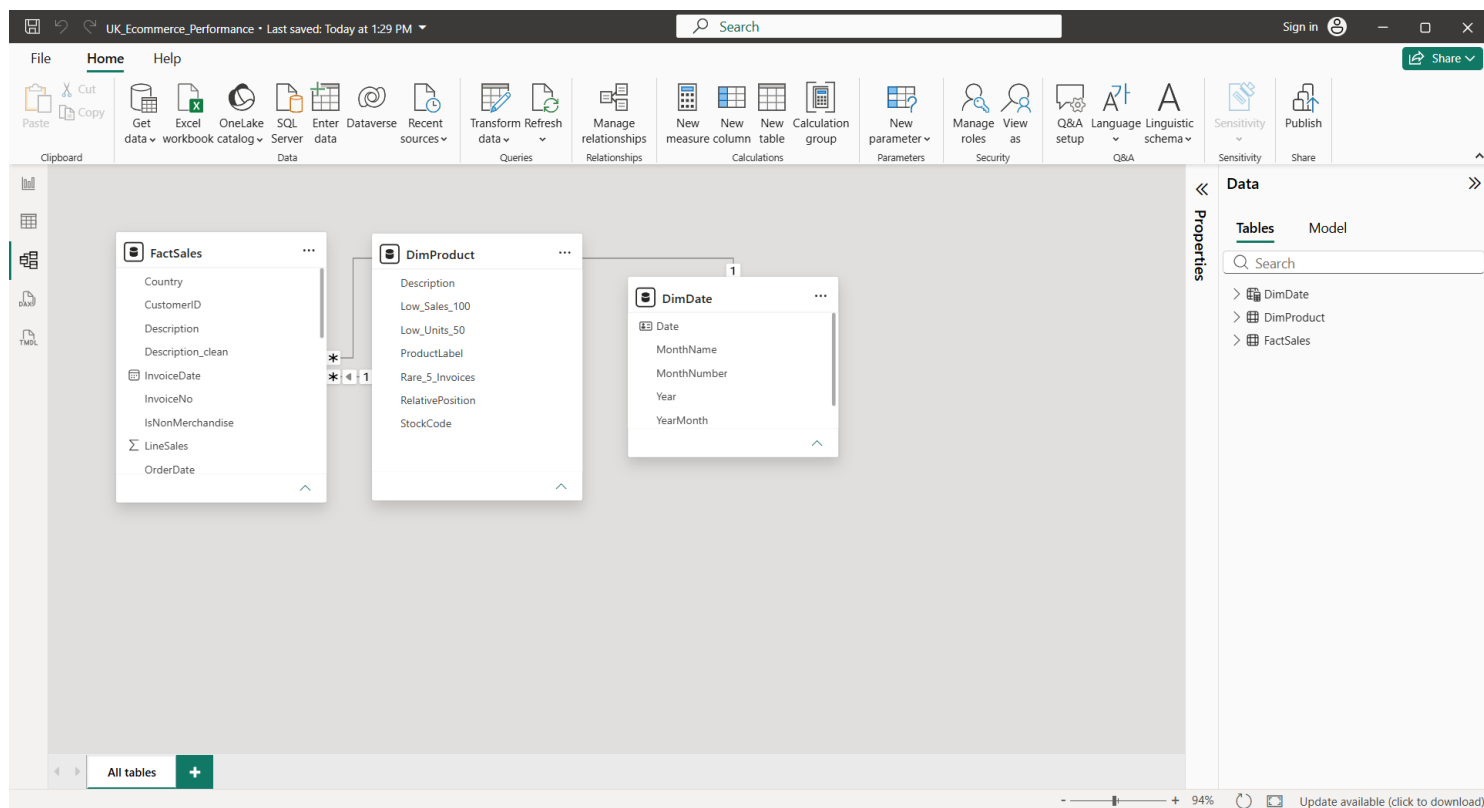
04 | Definitions and data quality

Coverage and source UK Online Retail dataset, December 1, 2010–December 9, 2011. Each source row represents an invoice line. December 2011 is incomplete.	Recorded positive value Sum of Quantity × UnitPrice for retained lines with positive quantity and price. Includes separately classified charges and accounting entries. Excludes negative records; it is not net revenue or profit.
Named-merchandise sales Recorded positive value excluding identified charges, accounting entries, ambiguous MANUAL entries, and missing descriptions.	Customer analysis Rankings and customer-value shares use identified customers only. Missing CustomerIDs remain in overall value totals.
Product portfolio groups Median-based sales/units groups calculated across the full study period. They are descriptive classifications, not measures of profitability.	Matching negative entries Positive invoices 581483 and 541431 have matching negative entries recorded 12 and 16 minutes later respectively. Their combined positive value is £245,653.20. Use the review-status slicer to inspect their effect.
Full-period observation November 2011 recorded the highest full-month positive value. Invoice count increased more than average invoice value, which declined from October. The data does not establish the cause.	Limitations Returns are excluded from the positive-value source. Missing IDs, exact-row deduplication, product classification, and partial-month coverage affect interpretation. No margin or inventory-cost data is available.

Key distinction. Positive recorded values exclude negative records and include classified accounting/charge entries; merchandise is a narrower view. Customer rankings exclude missing IDs. Product quadrants describe sales/units, not profitability.

Large-line investigation. Invoices 581483 and 541431 have matching negatives C581484 and C541433, recorded 12 and 16 minutes later. Combined positive value: £245,653.20 (2.31%). Matching is strong reversal evidence, not an explicit source linkage or a general return-matching method.

05 | Model and reproducibility



Model. FactSales holds invoice lines; DimDate and DimProduct provide one-to-many filtering. Agreed relationships are DimDate[Date] to FactSales[OrderDate] and DimProduct[StockCode] to FactSales[StockCode], active and single direction.

Refresh. Rerun Task 1, regenerate both CSVs, update local Source paths and refresh Power BI. Preserve InvoiceNo/StockCode as text; normalise stock-code case before product aggregation. Do not reuse the old exact product count after consolidation. See dax_measures.md for formulas, types and validation.

Recommendations, limits and validation

Evidence	Proposed action	Additional evidence needed
Higher November invoice activity	Review fulfilment and stock availability ahead of comparable busy periods	Multi-year demand, stockouts, capacity and lead times
Material value from leading customer accounts	Review repeat ordering and account relationships	Return-adjusted spending and retention history
Different product sales/units profiles	Review range and replenishment priorities	Margin, availability, product age and inventory cost
High Netherlands invoice average; nine customers	Investigate economics of existing international accounts	Customer concentration, logistics and repeat purchases
Two large lines with matching negatives	Inspect rankings using the review-status filter	Complete returns analysis before any net-sales claim

Sensitivity validation. With all countries and full actual coverage, selecting Other positive lines should produce £10,396,457.60 recorded positive value and £10,009,365.98 merchandise sales. These subtract only the two investigated lines; they are not net sales.

Limits. Missing IDs, exact-row deduplication assumptions, classification judgment and incomplete December affect interpretation. Costs, margins and complete net-sales measures are unavailable. Screenshots alone cannot verify slicer behaviour or internal DAX; inspect the PBIX for implementation audit.

Project files. Open report/UK_Ecommerce_Performance.pbix in Power BI Desktop for interaction. Place this PDF and dax_measures.md in docs/. GitHub offers the PBIX for download rather than rendering it live.

References. Dataset: kaggle.com/datasets/carrie1/ecommerce-data. Model guidance: learn.microsoft.com/en-us/power-bi/guidance/star-schema. Formula definitions and source refresh instructions: companion dax_measures.md.